

```
exercise_2.py x
1  ## Course 1: Natural Language Processing
2  ## Module 1: Intro to NLP in Python
3  ## Coding Exercise 2
4
5  from nltk.tokenize import word_tokenize
6
7  def common_words(sentences):
8      count_dict = {}
9      for sentence in sentences:
10         words = word_tokenize(sentence)
11         for word in words:
12             word = word.lower()
13             count_dict[word] = count_dict.get(word, 0) + 1
14         return sorted(count_dict.items(), key=lambda item: item[1], reverse=True)[:4]
15
16 many_sentences = ["How many blankets do you want?", "There are many dogs here.", "How come there a
17 print(common_words(many_sentences))
```

Collapse



Coding Exercise 2

Problem

Create the function `common_words` that takes a list of strings and returns the four most common words used in the list.

Expected Output

After creating the `common_words` function, pass it the following variable and print the return value.

```
many_sentences = ["How many blankets do you want?", "There are many dogs here.", "H
```

Click the `TRY IT` button to run the file and check your work. The expected output is shown below.

TRY IT

[('many', 4), ('there', 3), ('are', 3), ('how', 2)]

[('many', 4), ('there', 3), ('are', 3), ('how', 2)]

Submit your work for evaluation.

Here is one possible solution:

```
from nltk.tokenize import word_tokenize
from nltk import FreqDist

many_sentences = ["How many blankets do you want?", "There are many dogs

def common_words(data):
    tokens_by_sentence = [word_tokenize(sentence) for sentence in data]
    tokens = sum(tokens_by_sentence, [])
    lowercase_tokens = [token.lower() for token in tokens]
    return (FreqDist(lowercase_tokens).most_common(4))

print(common_words(many_sentences))
```

- Import `FreqDist` and `word_tokenize` from the `nltk` module
- Create the `common_words` function
- Use a list comprehension to tokenize each sentence in the list passed to the function
- Use the `sum` function to "flatten" the 2D list of tokens
- Convert all of the tokens into lowercase strings
- Use `FreqDist` to find the four most common words

Check It!

```
LAST RUN on 12/25/2023, 1:38:10 PM

[('many', 4), ('there', 3), ('are',
3), ('how', 2)]
test_1 (test_exercise_2.test_ex2) ...
ok
test_2 (test_exercise_2.test_ex2) ...
ok
test_3 (test_exercise_2.test_ex2) ...
ok
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```

Next